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# MaxionBench: Conformance-Gated Decision Audits for Agentic Retrieval Memory

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## Abstract

1        Agentic AI systems increasingly use retrieval stores as mutable operational mem-  
2        ory: agents insert evidence, query it under interaction-latency budgets, and pay  
3        downstream context costs for retrieved passages. Existing evaluations split these  
4        concerns across static information-retrieval benchmarks and vector-database per-  
5        formance tests. We present MaxionBench, a conformance-gated benchmark that  
6        evaluates retrieval infrastructure as a deployment decision audit rather than as a sin-  
7        gle leaderboard. A fixed adapter contract gates engines before scoring; three work-  
8        loads test static retrieval, streaming post-insert retrievability, and multi-evidence  
9        retrieval; and each decision card reports quality, a normalized context-cost proxy,  
10       max-row p99 latency, budget stability, objective sensitivity, and paired audits. On a  
11       commodity single-node evaluation of FAISS CPU, LanceDB in-process, pgvec-  
12       tor, and Qdrant, a strict 200 ms max-row p99 policy selects FAISS CPU with  
13       BAAI/bge-small-en-v1.5 for S1, S2, and S3. The margins reveal different  
14       mechanisms: S1 and S2 are cost-quality ties resolved by tail latency, while S3 has  
15       lower cost and higher evidence coverage than the nearest strict-cost alternative. Re-  
16       moving the p99 cap or optimizing quality changes selected configurations, paired  
17       audits show that an aggregate S3 pgvector edge is indistinguishable from FAISS  
18       under matched queries, and short B0 screening runs fail to preserve full-budget  
19       B2 top-1 decisions for S1 and S2. MaxionBench therefore reports auditable local  
20       deployment evidence, not universal vector-database rankings.

## 21    1 Introduction

22       Retrieval stores are becoming operational memory for agentic AI systems. Unlike static search  
23       applications, agents repeatedly insert, update, and query evidence while operating under tail-latency  
24       and context-window constraints. A retrieval stack can therefore pass a static relevance benchmark but  
25       still fail deployment: newly inserted evidence may not be retrievable in the agent’s top- $k$  context,  
26       max-row p99 latency may exceed an interaction budget, or a high-quality configuration may impose  
27       an unacceptable downstream token footprint. This paper studies whether a lightweight, commodity-  
28       hardware benchmark can produce auditable local deployment evidence for these retrieval-memory  
29       choices.

30       Existing evidence is fragmented across static relevance and systems performance. Information-  
31       retrieval benchmarks such as BEIR [Thakur et al., 2021] and HotpotQA [Yang et al., 2018] provide  
32       reusable corpora and judgments, but they do not specify adapter conformance, incremental write se-  
33       mantics, context-cost accounting, or deployment objectives. Vector-search and database benchmarks  
34       report latency, throughput, and recall surfaces, but those measurements are often separate from task  
35       relevance and post-insert retrievability. The resulting gap is a decision problem: an operator needs  
36       to know which retrieval configuration is viable under a stated objective, not which engine wins one  
37       isolated metric.

MaxionBench addresses this gap by treating benchmarking as a conformance-gated deployment decision audit. A configuration is reportable only after passing a fixed adapter contract. It is then evaluated on workloads that isolate static retrieval, streaming post-insert retrievability, and multi-evidence retrieval, and the final report exposes quality, a normalized context-cost proxy, max-row p99 latency, budget stability, objective sensitivity, and paired decision margins. The protocol is intentionally scoped to local single-node deployment evidence; its purpose is to make the decision assumptions explicit and auditable rather than to rank vector databases universally.

The archived evaluation supports four findings. First, under a strict 200 ms max-row p99 policy, FAISS CPU with BAAI/bge-small-en-v1.5 is the policy-selected strict-latency configuration for S1, S2, and S3. Second, this is not a universal ranking: removing the p99 cap changes S3 to LanceDB in-process, and quality-first objectives choose different configurations. Third, paired audits distinguish real margins from fragile aggregate differences: S1 and S2 are tail-latency tie-breaks, and the S3 pgvector-base aggregate edge is indistinguishable from FAISS in a 5,000-query matched audit against FAISS exact FlatIP. Fourth, short budget runs are not reliable final-decision substitutes: B0-to-B2 top-1 agreement is absent for S1 and S2, while S3 preserves top-1 despite low full-rank correlation.

The paper contributes:

1. A conformance-gated adapter protocol for retrieval infrastructure used as agentic memory, covering lifecycle operations, bulk and incremental writes, query and batch query, payload/vector updates, deletes, flush/commit behavior, index controls, and minimum stats fields.
2. Three MaxionBench workloads covering static retrieval, streaming post-insert top-10 retrievability, and multi-evidence retrieval under local embedding tiers.
3. A decision-audit reporting format that separates strict-latency, cost-only/no-p99, and quality-first objectives, then reports quality, normalized context-cost proxy, max-row p99 latency, budget stability, margins, and paired audits.
4. Empirical local findings showing that strict-latency, cost-only/no-p99, and quality-first objectives select different configurations; that aggregate quality edges can vanish under matched audits; and that short screening runs do not reliably preserve full-budget decisions.

The submitted artifact contains the benchmark code, archived run bundle, validation commands, conformance matrix, generated paper tables and figures, query-level paired-audit summaries, checksums, HOTPOTQA-MAXIONBENCH metadata, and a claim-evidence map.

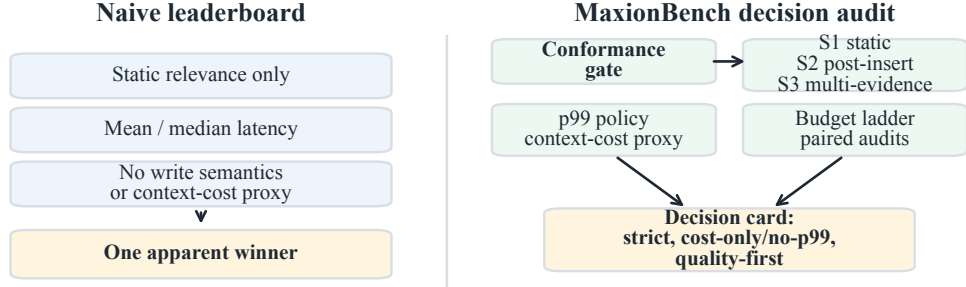
## 2 Benchmark design

### 2.1 Scope and systems under test

MaxionBench evaluates retrieval infrastructure rather than full agent task success. The benchmark is intentionally single-node: all experiments run on one commodity operator machine, and the paper does not make distributed-systems or GPU-scaling claims. The reportable engine set contains FAISS CPU [Johnson et al., 2019], LanceDB in-process [LanceDB, 2024], PostgreSQL with pgvector [pgvector contributors, 2024], and Qdrant [Qdrant contributors, 2024]. LanceDB service mode is listed but excluded from the paper matrix because the archived conformance table has no passing local service row.

MaxionBench is organized as a decision-audit pipeline rather than as a single leaderboard: a reported winner is valid only under a stated conformance gate, workload, objective, and run budget. The engine set is intended to cover representative local operating modes rather than exhaust the vector-database landscape: an exact in-process baseline, an embedded/local store, a PostgreSQL extension, and a service vector database. FAISS CPU should be interpreted as an exact in-process local baseline and deployment option, not as a feature-equivalent product database comparison. The adapter contract is the extension path for additional engines such as Milvus, Weaviate, Redis, Vespa, OpenSearch, or Chroma; absence from this local matrix should not be read as inferiority.

Every reportable engine must pass the same adapter contract before its runs can enter paper-facing tables. The contract covers lifecycle operations, bulk and incremental writes, query and batch query, vector and payload updates, delete semantics, flush or commit behavior, index controls, and minimum



A reportable winner is conditional on conformance, workload, objective, budget, and audit margins.

Figure 1: From leaderboard evaluation to decision audit. MaxionBench makes a deployment choice reportable only after conformance, workload evidence, p99 policy, context-cost accounting, budget stability, and paired margins are all visible.

Table 1: MaxionBench workload summary. S1/S2 counts come from the processed local BEIR-format bundles under the 50K-document paper cap; S3 counts come from the HotpotQA-MaxionBench manifest. Cov@10 denotes evidence\_coverage@10, and all-hit@10 denotes all\_evidence\_hit@10.

| Workload | Assets                                                      | Corpus                                                    | Queries / events                                           | Labels                                    | Decision role                                                 | Metrics                       |
|----------|-------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------|-------------------------------|
| S1       | BEIR-format<br>SciFact + FiQA                               | 50,000 docs                                               | 894 queries                                                | 1,685 qrels                               | static relevance and tail-latency baseline                    | nDCG@10                       |
| S2       | 50K BEIR<br>background +<br>CIRAG-500 stream                | 50,000<br>background +<br>500 inserted<br>stream passages | 894 static queries;<br>500 stream events                   | 1,685 static<br>qrels; 500<br>event qrels | post-insert retrievability<br>diagnostic under a write stream | nDCG@10;<br>post-insert@1s/5s |
| S3       | HOTPOTQA-<br>MAXIONBENCH<br>from HotpotQA<br>dev distractor | 66,635 docs                                               | 7,405 questions;<br>5,000-query<br>matched-audit<br>subset | 14,810 qrels                              | multi-evidence and<br>context-cost decision stress            | Cov@10; all-hit@10            |

stats fields. The conformance suite checks CRUD correctness, post-flush visibility, filter correctness, update semantics, delete semantics, and streaming-memory post-insert retrievability semantics. This gate prevents a benchmark table from silently mixing engines with different unsupported behavior and is the main reason the paper treats missing conformance as an exclusion rather than a bad score.

## 2.2 Workloads

MaxionBench uses three workloads designed to isolate retrieval-infrastructure behavior that matters to agentic applications. S1 is single-hop retrieval over SciFact and FiQA through local BEIR-format bundles [Thakur et al., 2021, Wadden et al., 2020, Maia et al., 2018]; its primary quality metric is nDCG@10. S2 combines a deterministic 50K-passages static background from the same source family with a CIRAG-500 event stream [Facebook Research, 2024]; it reports static nDCG@10 and post-insert top-10 retrievability probes at one and five seconds after insert acknowledgment. S3 uses HOTPOTQA-MAXIONBENCH, a frozen preprocessing of the HotpotQA dev distractor split, and measures whether the retrieved top-10 contains the supporting evidence set with evidence\_coverage@10. HOTPOTQA-MAXIONBENCH contains 7,405 questions; the paper-facing matched S3 audit uses the configured 5,000-query subset recorded by the run configuration.

The workloads use fixed concurrency pins to keep the protocol small enough for a one-day local run. S1 and S3 evaluate read clients {1, 4, 8}. S2 uses eight read clients and two serialized write clients, then probes each inserted event passage at  $T + 1s$  and  $T + 5s$ , where  $T$  is the time at which the adapter’s insert plus flush path returns. Each workload is evaluated with two local embedding tiers, BAAI/bge-small-en-v1.5 and BAAI/bge-base-en-v1.5 [Beijing Academy of Artificial Intelligence, 2023]. We write bge-small and bge-base after first mention.

## 2.3 Metrics and decisions

MaxionBench reports quality, tail latency, throughput, a normalized context-cost proxy, and stability across budget levels. Latency is measured on precomputed query vectors and times adapter .query plus top-k materialization, including service/container overhead inside the adapter call when applicable; offline embedding is excluded from latency and included only in the context-cost proxy. We rank latency by max-row p99: the maximum observed p99 across the configured B2 read-client or read/write-client rows and repeats for a candidate. This is a conservative local audit statistic, not a production latency guarantee or a hardware-population interval. For S1 and S2, quality is nDCG@10. For S3, quality is

$$\text{evidence\_coverage@k}(q) = \frac{|E(q) \cap R_k(q)|}{|E(q)|},$$

where  $E(q)$  is the gold evidence set and  $R_k(q)$  is the retrieved top- $k$  set. For S2,  $\text{post\_insert\_hit@10}$ ,  $\Delta$  is one when the newly inserted supporting passage appears in  $R_{10}(q, T + \Delta)$  for  $\Delta \in \{1s, 5s\}$  after the adapter’s insert-plus-flush path returns. This is an agent-facing retrievability metric and not a direct measurement of low-level index visibility. For S2, the archived strict-latency rule gates on static nDCG@10 and reports  $\text{post\_insert\_hit@10, 1s}$  and  $\text{post\_insert\_hit@10, 5s}$  as agent-facing write/retrieve diagnostics. We therefore interpret S2 deployment choices as static-quality/cost/p99 decisions accompanied by post-insert retrievability evidence, not as claims about low-level index visibility.

Within each workload and configured subset, all reportable engines evaluate the same quality query or event IDs; latency and throughput observation counts may differ because they are measured in timed windows. Paired audits join explicitly by query ID or event index. Quality differences across engines therefore reflect configured search behavior, tie handling, filtering/update semantics, and top-k materialization under the adapter contract rather than differences in the embedding model itself. In the archived MaxionBench matrix, the FAISS CPU rows have empty `index_params` and therefore use the adapter’s exact flat index; recorded `hnsw_ef` sweep values are not consumed by that flat-index path.

The deployment rule is deliberately simple. A strict-latency survivor must pass the conformance gate, satisfy the workload quality floor recorded in the run configuration, and remain below the stated max-row p99 cap. For each workload, engine, and embedding tier, these survivors are ranked by the normalized context-cost proxy

$$\text{task\_cost\_est} = C_{\text{retrieval}} + C_{\text{embedding}} + c_{\text{llm\_in}} \cdot N_{\text{retrieved\_input\_tokens}},$$

with lower max-row p99 and higher throughput used as tie-breaks. The artifact field is named `task_cost_est`, but the paper interprets it only as a normalized context-token cost proxy: it is not a cloud-dollar estimate. In the archived local run,  $C_{\text{embedding}} = 0$  for offline embeddings and  $c_{\text{llm\_in}} = 0.15$  is a project-defined token coefficient. The strict-latency objective selects the lowest-cost survivor under a 200 ms max-row p99 cap, which we treat as an example operator policy for an agent-interaction tail-latency budget rather than a universal standard. The report also exposes cost-only/no-p99 and quality-first objectives because those plausible operator choices can select different configurations.

**Decision-card protocol.** Given engines  $E$ , embedding tiers  $M$ , workloads  $W$ , budget levels  $B$ , a p99 cap  $\tau$ , and objectives  $O$ , MaxionBench runs the following audit:

1. run adapter conformance checks for each engine and exclude non-conforming rows;
2. evaluate quality, max-row p99, throughput, and context-cost proxy for each workload and embedding tier;
3. rank surviving configurations under each stated objective and record strict-latency, cost-only/no-p99, and quality-first choices;
4. compute B0/B1/B2 stability statistics and run paired audits for high-risk aggregate margins;
5. emit a decision card with choices, margins, stability, conformance status, and caveats.

## 2.4 Budget ladder

The benchmark uses three budget levels to test whether decisions are stable under shorter local runs. B0 uses 10 seconds of warmup, 10 seconds of measurement, and one repeat. B1 uses 15 seconds of warmup, 30 seconds of measurement, and one repeat. B2 uses 30 seconds of warmup, 60 seconds of measurement, and two repeats. Stability is reported as Spearman rank correlation, top-1 agreement,

and top-2 agreement between budget-level rankings. This design treats instability as a result to report, not as a failure to hide.

### 3 Related work

Information-retrieval benchmarks provide the relevance backbone for this work. BEIR [Thakur et al., 2021], SciFact [Wadden et al., 2020], FiQA [Maia et al., 2018], and HotpotQA [Yang et al., 2018] supply reusable corpora and judgments, but they do not specify retrieval-engine conformance, write/retrieve probes, context-cost estimates, or deployment decision rules. MaxionBench reuses labeled retrieval evidence while adding the systems and decision-audit layer needed for local infrastructure choices.

Vector-search benchmarks usually emphasize algorithmic recall, throughput, indexing configuration, and operating mode. ANN-Benchmarks [Aumüller et al., 2020], the NeurIPS billion-scale ANN challenge [Simhadri et al., 2021], and vector-database suites such as VectorDBBench [Zilliz, 2023] expose important search-performance surfaces. These benchmarks are complementary to MaxionBench: they characterize ANN or product performance, whereas MaxionBench fixes an agent-facing adapter contract and ties configured search behavior to retrieval quality, post-insert top-10 retrievability, context-token cost, and deployment objectives.

Benchmark methodology also motivates the reporting style. MLPerf Inference [Reddi et al., 2020] emphasizes rules and reproducible system comparisons; HELM [Liang et al., 2023] emphasizes multi-metric transparency; datasheets [Gebru et al., 2021] and Croissant metadata [MLCommons, 2024] motivate explicit dataset documentation; and tail-latency work motivates high-percentile latency reporting [Dean and Barroso, 2013]. MaxionBench applies this emphasis on conformance, metadata, and multi-metric trade-offs to retrieval stores used as local agent memory.

RAG and agent-memory evaluations study whether systems retrieve and use evidence in task contexts. RAGAS [Es et al., 2024], CRAG [Facebook Research, 2024], RAGBench [Friel et al., 2024], AgentBench [Liu et al., 2023], MemGPT [Packer et al., 2023], LongMemEval [Wu et al., 2024], and LoCoMo [Maharana et al., 2024] evaluate RAG quality, agent behavior, or long-term conversational memory. MaxionBench focuses below full agent task success on the retrieval infrastructure layer: can inserted evidence be retrieved, can p99 stay within budget, and does the deployment decision survive objective and budget changes?

## 4 Experiments

### 4.1 Artifact audit

The paper is built from the archived bundle `results/20260429T033427Z`. The archive contains the run artifacts, generated figures, HOTPOTQA-MAXIONBENCH manifest and checksums, conformance matrix, and operator command documents. A strict-schema validation pass over the archived MaxionBench runs checked 72 run directories and returned zero errors. Appendix Table 14 reports passing conformance and behavior-card coverage for FAISS CPU, LanceDB in-process, pgvector, and Qdrant; LanceDB service is excluded because it has no conformance row.

The artifact is intended to make each paper claim traceable. It contains the benchmark code, archived run bundle, validation commands, conformance matrix, generated tables and figures, query-level paired-audit summaries, checksums, HOTPOTQA-MAXIONBENCH Croissant-style metadata [MLCommons, 2024], and a claim-evidence map. All main results below should therefore be read as local single-node deployment evidence under the recorded configs, not as universal engine rankings.

### 4.2 Decision audit results

We organize results around four deployment-audit questions rather than a single leaderboard ranking. First, under the default 200 ms max-row p99 policy, FAISS CPU with bge-small is the policy-selected strict-latency configuration for all three workloads. Second, this decision is objective-sensitive: removing the p99 cap or optimizing quality changes selected configurations. Third, paired audits separate clear margins from aggregate near-ties. Fourth, budget stability is workload-dependent, so short screening runs cannot always replace B2 decisions.

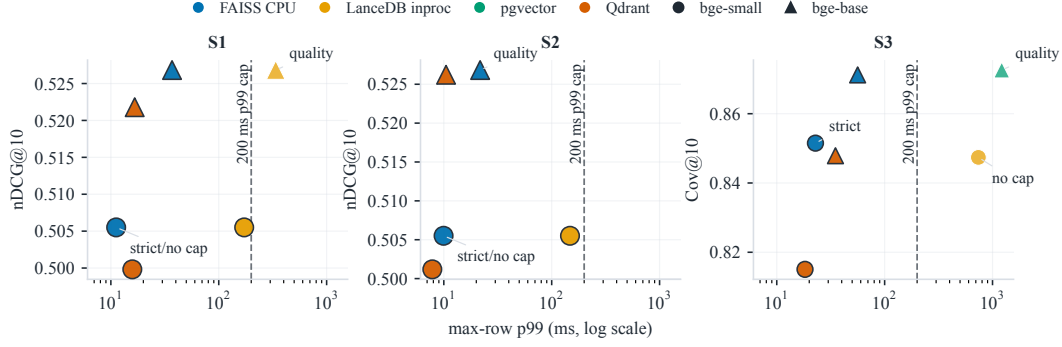


Figure 2: Decision surface for the B2 audit. The strict 200 ms max-row p99 policy selects FAISS CPU/bge-small for S1–S3, but the no-p99 and quality-first markers show that the selection is conditional on objective and policy. Each panel uses a workload-specific zoomed y-axis.

Table 2: Strict-latency deployment cards. Quality and S2 post-insert top-10 retrievability report 95% confidence intervals; p99 max is the maximum observed p99 across configured B2 client rows and repeats for the selected configuration.

| Workload          | Engine    | Emb.      | Quality                     | post_insert_hit@10,5s | p99 max | Stability     |
|-------------------|-----------|-----------|-----------------------------|-----------------------|---------|---------------|
| S1 single-hop     | faiss-cpu | bge-small | nDCG@10 0.506 (0.494–0.516) | –                     | 11.2 ms | top-1 changed |
| S2 streaming      | faiss-cpu | bge-small | nDCG@10 0.506 (0.487–0.524) | 0.830 (0.805–0.852)   | 10.0 ms | top-1 changed |
| S3 multi-evidence | faiss-cpu | bge-small | Cov@10 0.852 (0.849–0.854)  | –                     | 22.8 ms | top-1 stable  |

Table 2 is a strict-latency deployment-card summary. Under the 200 ms max-row p99 threshold, all three workloads select FAISS CPU with BAAI/bge-small-en-v1.5. The FAISS rows are exact FlatIP local-baseline rows, while service engines include their adapter and service overhead; the table therefore supports a local deployment decision, not a distributed product ranking. The S1 selected row reaches nDCG@10 0.506 with a 95% confidence interval of 0.494–0.516 and max-row p99 11.2 ms. S2 selects the same engine and embedding, with nDCG@10 0.506, post\_insert\_hit@10,5s 0.830, and max-row p99 10.0 ms. S3 selects FAISS CPU with BAAI/bge-small-en-v1.5 at evidence\_coverage@10 0.852 and max-row p99 22.8 ms.

S2 should be interpreted as agent-facing post-insert top-10 retrievability, not low-level index visibility. S1 and the static-retrieval component of S2 use the same 894-query BEIR-format evaluation pool under the 50K-document cap, which is why their static nDCG@10 and packed-token summaries can match. S2 adds CRAG-500 write/retrieve probes after the adapter insert-plus-flush path returns. Across two B2 strict-choice repeats with 500 events each, 830 of 1,000 post-insert observations are retrieved by the 1s probe; no additional observations recover by 5s, leaving 170 of 1,000 absent at the 5s probe. Appendix Figure 6, Appendix Table 12, and Appendix Table 13 report the fixed-probe diagnostics and example event/query/document IDs.

The benchmark-design ablation is reported in Appendix Table 9. Its main implication is that removing any audit component changes the interpretation: the p99 gate separates strict-latency decisions from cost-only/no-p99 choices, objective separation prevents quality-first rows from being conflated with strict-latency rows, budget reporting exposes unstable S1/S2 screening runs, paired audits prevent aggregate near-ties from being overclaimed, and conformance filtering keeps unsupported service rows out of paper-facing tables.

Table 3 and Figure 2 show objective sensitivity directly. The strict-latency and cost-only/no-p99 objectives agree for S1 and S2, but S3 shifts to LanceDB in-process with BAAI/bge-small-en-v1.5 when the p99 cap is removed. Quality-first decisions differ more strongly: LanceDB in-process with BAAI/bge-base-en-v1.5 leads S1, FAISS CPU with BAAI/bge-base-en-v1.5 leads S2, and the archived aggregate rule selects pgvector with BAAI/bge-base-en-v1.5 for S3. The point is not that any one 200 ms threshold is universal; the point is that MaxionBench makes the policy explicit and auditable.

The S3 quality-first margin is the highest-risk aggregate claim, so we audit it with matched queries. Using all B2 rows, pgvector-base exceeds FAISS-base by only 0.00016 evidence\_coverage@10

Table 3: Deployment decisions under strict-latency, cost-only/no-p99, and quality-first objectives. Strict latency uses a 200 ms max-row p99 threshold; the S3 quality-first aggregate choice is treated as a matched-audit tie case in the text.

| Workload          | Strict p99            | Cost-only/no-p99           | Quality-first                     | Spearman B0-B2 | Stability     |
|-------------------|-----------------------|----------------------------|-----------------------------------|----------------|---------------|
| S1 single-hop     | faiss-cpu / bge-small | faiss-cpu / bge-small      | lancedb-inproc / bge-base         | 0.79           | top-1 changed |
| S2 streaming      | faiss-cpu / bge-small | faiss-cpu / bge-small      | faiss-cpu / bge-base              | 0.43           | top-1 changed |
| S3 multi-evidence | faiss-cpu / bge-small | lancedb-inproc / bge-small | pgvector / bge-base (matched tie) | 0.31           | top-1 stable  |

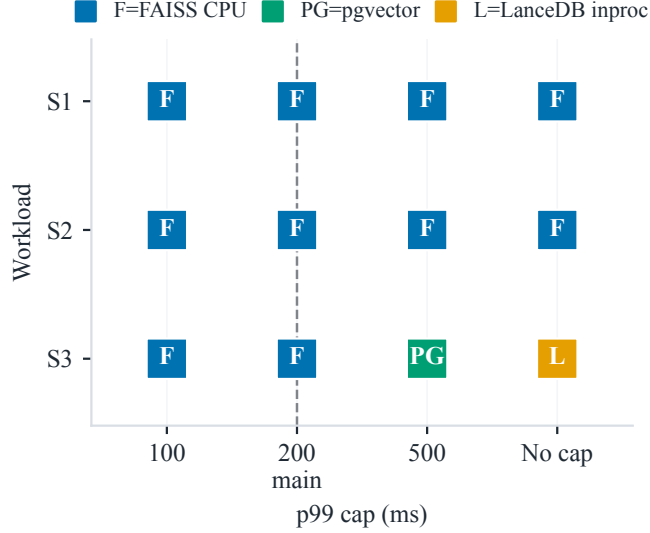


Figure 3: S3 is sensitive to relaxing the p99 policy, while S1 and S2 keep the same policy-selected engine across the reported thresholds. This threshold sweep supports the scope claim that Maxion-Bench reports conditional local decisions rather than universal rankings.

233 with an aggregate-row bootstrap interval of -0.00280 to 0.00304. The matched audit in Figure 4  
 234 removes even that apparent edge: pgvector IVF32 and IVF64 are each 0.0001 evidence\_coverage@10  
 235 below FAISS exact FlatIP on the 5,000-query matched-audit subset, with paired intervals including  
 236 zero. Appendix Table 17 gives the numeric rows, and Appendix Table 18 records the stricter binary  
 237 all\_evidence\_hit@10 audit.

238 Appendix Table 11 makes the strict-latency decision margins explicit. For S1 and S2, FAISS  
 239 CPU with BAAI/bge-small-en-v1.5 is not a relevance-quality win over the closest LanceDB  
 240 in-process candidate; it is a cost and quality tie that the deployment rule resolves by lower max-row  
 241 p99 and higher throughput. For S3, the nearest strict-cost alternative is not a tie: Qdrant with  
 242 BAAI/bge-small-en-v1.5 is more expensive and lower quality, even though its max-row p99  
 243 is 4.6 ms lower in the repeat-aware table. The highest-quality strict-latency S3 option remains  
 244 FAISS CPU with BAAI/bge-base-en-v1.5, which improves evidence\_coverage@10 by 0.020 over  
 245 BAAI/bge-small-en-v1.5 but increases the normalized context-cost proxy by 0.704 and max-row  
 246 p99 by 33.4 ms on the B2 audit rows.

247 The normalized context-cost proxy is stable under the coefficient sensitivity tested in the artifact.  
 248 Appendix Table 5 rescales only  $c_{llm\_in}$  from  $0.1\times$  to  $10\times$  and leaves all strict choices unchanged:  
 249 S1/S2 have identical retrieved-token counts for the strict choice and nearest strict-cost competitor,  
 250 while S3 FAISS/bge-small retrieves slightly fewer tokens than Qdrant/bge-small. Appendix Table 6  
 251 reports p50, p95, mean p99, and the min-max p99 range over archived B2 run rows; these are  
 252 conditional on the fixed local run and should not be read as population-level latency intervals.

253 The budget analysis qualifies how much evidence is enough. B0-to-B2 top-1 agreement is zero for S1  
 254 and S2, with Spearman correlations 0.79 and 0.43, respectively. S3 has top-1 agreement of one but  
 255 Spearman correlation 0.31, meaning that the selected strict-latency row is stable while lower-ranked  
 256 alternatives remain noisy. The benchmark can therefore produce actionable local decisions, but the  
 257 decision card must include the budget level and stability statistics used to reach them.

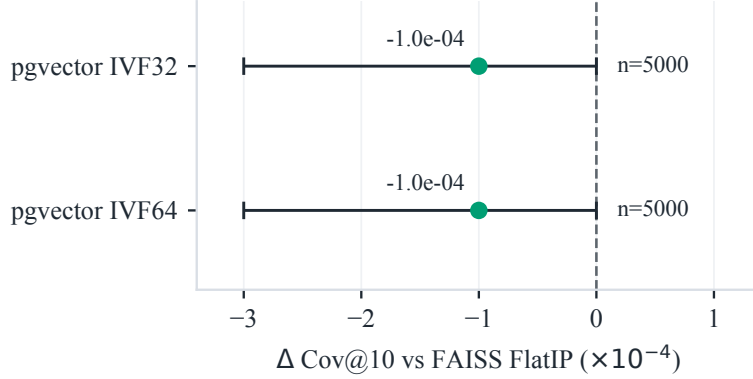


Figure 4: Matched S3 paired audit. Against FAISS exact FlatIP on the same 5,000 bge-base queries, pgvector IVF32 and IVF64 have paired evidence\_coverage@10 deltas whose 95% intervals include zero, so the aggregate pgvector quality-first row is indistinguishable under matched queries.

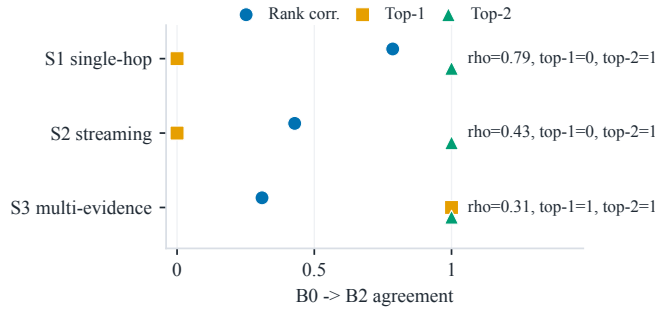


Figure 5: Budget stability is workload-dependent. B0 screening fails to preserve B2 top-1 choices for S1 and S2, while S3 preserves top-1 despite low full-rank Spearman correlation.

### 258 4.3 Targeted replication checks

259 Targeted same-machine reruns support the decision-level conclusions but not exact tail-latency  
 260 reproduction. Appendix Table 8 shows that the strict FAISS CPU/bge-small choices reproduce the  
 261 relevant quality values and remain under the 200 ms policy; the S3 bge-base rerun and Figure 4  
 262 support the objective-sensitivity story; and Appendix Table 20 reports a larger FAISS-versus-Qdrant  
 263 S2 paired check with no hidden Qdrant quality or post-insert retrievability win. Exact repeat numbers  
 264 remain in the appendix because p99 is local-run evidence rather than a hardware-population interval.

## 265 5 Limitations and broader impact

266 MaxionBench is a bounded benchmark, not a claim that commodity hardware can substitute for  
 267 production-scale evaluation. The experiments run on one local operator machine, so the results do  
 268 not characterize distributed retrieval, GPU indexing, cloud variability, second-machine portability, or  
 269 multi-tenant production workloads. S3 uses HOTPOTQA-MAXIONBENCH, a frozen paragraph-  
 270 level proxy for full-Wikipedia multi-evidence retrieval, and S2 measures top-k retrievability after  
 271 inserts rather than lower-level index visibility. Engine-level quality differences are configured-search  
 272 differences under the MaxionBench adapter settings, not inherent database quality. The S3 paired  
 273 audit uses FAISS exact FlatIP as a local reference for the bge-base comparison, but the archived  
 274 matrix does not sweep higher service-engine settings such as 128/256 probes or ef values. These  
 275 choices keep the benchmark reproducible and one-day executable while narrowing the scope of the  
 276 claims.

277 The uncertainty estimates are conditional on this fixed benchmark setting. Confidence intervals are  
 278 query- or event-level summaries, not hardware-population intervals, and max-row p99 latency is not a



279 production tail-latency guarantee. S1 and S2 change top-1 recommendations between B0 and B2, so  
280 a screening run is not enough to make a final deployment decision for those workloads; S3 keeps the  
281 same top-1 recommendation but has low full-rank correlation. The broader impact is methodological:  
282 a reproducible benchmark can help operators choose lower-cost, lower-latency retrieval infrastructure  
283 without hiding quality or post-insert retrievability tradeoffs, but readers may overread the winner as a  
284 universal recommendation. The manuscript mitigates that risk by reporting exclusions, limitations,  
285 budget stability, and objective sensitivity directly.

## 286 **6 Conclusion**

287 MaxionBench frames retrieval infrastructure for agentic AI as an auditable deployment decision  
288 rather than a leaderboard. The key idea is to report a configuration only after it passes a fixed  
289 adapter contract, then expose workload quality, post-insert top-10 retrievability, max-row p99 latency,  
290 normalized context-cost proxy, budget stability, objective sensitivity, and paired margins in one  
291 decision card. The archived local evidence shows why that audit matters: the strict 200 ms policy  
292 selects FAISS CPU/bge-small for S1–S3, but alternative objectives change decisions, S1/S2 are tail-  
293 latency tie-breaks, the S3 pgvector aggregate edge is indistinguishable under matched queries, and  
294 short B0 runs do not preserve all B2 decisions. The supported conclusion is bounded but actionable:  
295 local retrieval-memory deployment choices are more credible when their conformance, objective,  
296 budget, and decision margins are explicit.

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## A Reproducibility details

The MaxionBench workflow is run from the repository root. The main commands are:

```
maxionbench workflow setup --json
maxionbench workflow data --json
maxionbench submit --budget b0 --json
maxionbench submit --budget b1 --json
maxionbench submit --budget b2 --json
maxionbench workflow finalize --json
```

The archived bundle used by this manuscript is `results/20260429T033427Z`. It contains 972 run-artifact files, 42 generated figure files, 27 HOTPOTQA-MAXIONBENCH files, and 27 conformance files. The strict-schema validation command used during manuscript preparation was:

```
python -m maxionbench.cli validate --input artifacts/runs/portable \
--strict-schema --json
```

It returned `"pass": true, "error_count": 0, and "run_dirs_checked": 72`.

The archived MaxionBench matrix contains 72 `results.parquet` files, including 24 B2 result files with matching `run_metadata.json` files. The public submission records a single-node CPU-only local runtime profile; host, OS, username, and machine-local path details are intentionally omitted from the repository copy. No accelerator-backed benchmark row is reported. HOTPOTQA-MAXIONBENCH contains 66,635 documents, 7,405 questions, and 14,810 qrels; the verified manifest fields are `doc_count=66635`, `question_count=7405`, and `qrel_count=14810`. The archived S3 run configuration records `d4_max_queries: 5000` and `s3_max_events: 5000`; these fields define the matched-audit subset cap and are not the full HOTPOTQA-MAXIONBENCH question count. The HotpotQA-MaxionBench checksum manifest stores SHA-256 checksums for the corpus, queries, qrels, metadata, and manifest.

The reviewer artifact also includes an artifact card, a local readiness verifier, and Croissant-style metadata for HOTPOTQA-MAXIONBENCH. The verifier checks that the manuscript, archive manifest, paper-facing tables, paired-audit summaries, and metadata files are present and internally consistent; the Croissant metadata records the corpus, query, and qrel files, their intended evaluation use, and the license/provenance boundary inherited from HotpotQA.

## B Audit tables

Table 4: Normalized context-cost proxy terms and deterministic value sources. The archived formula is  $task\_cost\_est = rhu\_h / \max(measured\_requests, 1) + 0.0 + c\_llm\_in \times avg\_retrieved\_input\_tokens$ , with  $c\_llm\_in = 0.15$  in the resolved run configs and metadata; it is not a cloud-dollar estimate.

| Term                        | Meaning                                                                     | Unit                                      | Value source                                                            |
|-----------------------------|-----------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------|
| $C\_retrieval$              | Per-query retrieval cost estimate                                           | project-defined normalized cost           | $retrieval\_cost\_est = rhu\_h / \max(measured\_requests, 1)$           |
| $C\_embedding$              | Per-query embedding cost estimate                                           | project-defined normalized cost           | $embedding\_cost\_est = 0.0$ for archived offline local embeddings      |
| $c\_llm\_in$                | Input-token cost coefficient                                                | project-defined normalized cost per token | 0.15 from RunConfig $c\_llm\_in$                                        |
| $Nretrieved\_input\_tokens$ | Retrieved context tokens packed downstream                                  | tokens/query                              | $avg\_retrieved\_input\_tokens$ measured from top-k outputs             |
| $task\_cost\_est$           | $C\_retrieval + C\_embedding + c\_llm\_in \times Nretrieved\_input\_tokens$ | normalized context-cost/query             | $retrieval\_cost\_est + embedding\_cost\_est + llm\_context\_cost\_est$ |

Table 5: Strict-choice cost sensitivity to input-token cost multipliers. The  $0.1\times$ ,  $1\times$ , and  $10\times$  rows rescale only  $c\_llm\_in$ ; strict choices do not change because the strict choice and nearest strict-cost competitor have identical S1/S2 token counts and lower S3 token count for FAISS CPU.

| Workload          | Role                           | Candidate                  | Mult. | Tokens | Cost     | Changes? | Selected              |
|-------------------|--------------------------------|----------------------------|-------|--------|----------|----------|-----------------------|
| S1 single-hop     | strict choice                  | faiss-cpu / bge-small      | 0.1   | 1752   | 26.276   | no       | faiss-cpu / bge-small |
| S1 single-hop     | nearest strict-cost competitor | lancedb-inproc / bge-small | 0.1   | 1752   | 26.276   | no       | faiss-cpu / bge-small |
| S1 single-hop     | strict choice                  | faiss-cpu / bge-small      | 1.0   | 1752   | 262.759  | no       | faiss-cpu / bge-small |
| S1 single-hop     | nearest strict-cost competitor | lancedb-inproc / bge-small | 1.0   | 1752   | 262.759  | no       | faiss-cpu / bge-small |
| S1 single-hop     | strict choice                  | faiss-cpu / bge-small      | 10.0  | 1752   | 2627.592 | no       | faiss-cpu / bge-small |
| S1 single-hop     | nearest strict-cost competitor | lancedb-inproc / bge-small | 10.0  | 1752   | 2627.592 | no       | faiss-cpu / bge-small |
| S2 streaming      | strict choice                  | faiss-cpu / bge-small      | 0.1   | 1752   | 26.276   | no       | faiss-cpu / bge-small |
| S2 streaming      | nearest strict-cost competitor | lancedb-inproc / bge-small | 0.1   | 1752   | 26.276   | no       | faiss-cpu / bge-small |
| S2 streaming      | strict choice                  | faiss-cpu / bge-small      | 1.0   | 1752   | 262.759  | no       | faiss-cpu / bge-small |
| S2 streaming      | nearest strict-cost competitor | lancedb-inproc / bge-small | 1.0   | 1752   | 262.759  | no       | faiss-cpu / bge-small |
| S2 streaming      | strict choice                  | faiss-cpu / bge-small      | 10.0  | 1752   | 2627.592 | no       | faiss-cpu / bge-small |
| S2 streaming      | nearest strict-cost competitor | lancedb-inproc / bge-small | 10.0  | 1752   | 2627.592 | no       | faiss-cpu / bge-small |
| S3 multi-evidence | strict choice                  | faiss-cpu / bge-small      | 0.1   | 862    | 12.932   | no       | faiss-cpu / bge-small |
| S3 multi-evidence | nearest strict-cost competitor | qdrant / bge-small         | 0.1   | 866    | 12.984   | no       | faiss-cpu / bge-small |
| S3 multi-evidence | strict choice                  | faiss-cpu / bge-small      | 1.0   | 862    | 129.325  | no       | faiss-cpu / bge-small |
| S3 multi-evidence | nearest strict-cost competitor | qdrant / bge-small         | 1.0   | 866    | 129.835  | no       | faiss-cpu / bge-small |
| S3 multi-evidence | strict choice                  | faiss-cpu / bge-small      | 10.0  | 862    | 1293.249 | no       | faiss-cpu / bge-small |
| S3 multi-evidence | nearest strict-cost competitor | qdrant / bge-small         | 10.0  | 866    | 1298.351 | no       | faiss-cpu / bge-small |

Table 6: Latency distributions for strict choices and objective-sensitivity rows. Boundary: pre-computed query vector; timed adapter.query plus top-k materialization, including service/container overhead inside the adapter call when applicable; offline embedding is excluded. The p99 range is the min–max over archived B2 run rows for the listed configuration.

| Workload          | Role                    | Engine         | Embedding | R/W clients         | p50   | p95   | p99   | p99 range    | Rows/rep | Obs.  |
|-------------------|-------------------------|----------------|-----------|---------------------|-------|-------|-------|--------------|----------|-------|
| S1 single-hop     | strict choice           | faiss-cpu      | bge-small | 1 / 0, 4 / 0, 8 / 0 | 3.8   | 5.9   | 6.3   | 1.5–11.2     | 6/2      | 5364  |
| S1 single-hop     | cost-only/no-p99 choice | faiss-cpu      | bge-small | 1 / 0, 4 / 0, 8 / 0 | 3.8   | 5.9   | 6.3   | 1.5–11.2     | 6/2      | 5364  |
| S1 single-hop     | quality-first choice    | lancedb-inproc | bge-base  | 1 / 0, 4 / 0, 8 / 0 | 125.3 | 146.1 | 171.9 | 24.6–337.4   | 6/2      | 5364  |
| S2 streaming      | strict choice           | faiss-cpu      | bge-small | 8 / 2               | 5.7   | 9.8   | 10.0  | 9.9–10.0     | 2/2      | 1788  |
| S2 streaming      | cost-only/no-p99 choice | faiss-cpu      | bge-small | 8 / 2               | 5.7   | 9.8   | 10.0  | 9.9–10.0     | 2/2      | 1788  |
| S2 streaming      | quality-first choice    | faiss-cpu      | bge-base  | 8 / 2               | 12.0  | 21.1  | 21.7  | 21.4–21.7    | 2/2      | 1788  |
| S3 multi-evidence | strict choice           | faiss-cpu      | bge-small | 1 / 0, 4 / 0, 8 / 0 | 5.3   | 8.7   | 11.6  | 2.4–22.8     | 6/2      | 30000 |
| S3 multi-evidence | cost-only/no-p99 choice | lancedb-inproc | bge-small | 1 / 0, 4 / 0, 8 / 0 | 203.9 | 276.7 | 396.3 | 54.8–739.6   | 6/2      | 7469  |
| S3 multi-evidence | quality-first choice    | pgvector       | bge-base  | 1 / 0, 4 / 0, 8 / 0 | 350.2 | 540.5 | 715.2 | 268.3–1217.3 | 6/2      | 2827  |

Table 7: Latency distributions for nearest strict-cost competitors. Boundary: precomputed query vector; timed adapter.query plus top-k materialization, including service/container overhead inside the adapter call when applicable; offline embedding is excluded. The p99 range is the min–max over archived B2 run rows for the listed configuration.

| Workload          | Role                           | Engine         | Embedding | R/W clients         | p50  | p95  | p99   | p99 range  | Rows/rep | Obs.  |
|-------------------|--------------------------------|----------------|-----------|---------------------|------|------|-------|------------|----------|-------|
| S1 single-hop     | nearest strict-cost competitor | lancedb-inproc | bge-small | 1 / 0, 4 / 0, 8 / 0 | 66.1 | 77.9 | 96.4  | 17.1–172.1 | 6/2      | 5364  |
| S2 streaming      | nearest strict-cost competitor | lancedb-inproc | bge-small | 8 / 2               | 78.3 | 91.9 | 147.8 | 96.5–147.8 | 2/2      | 1788  |
| S3 multi-evidence | nearest strict-cost competitor | qdrant         | bge-small | 1 / 0, 4 / 0, 8 / 0 | 7.2  | 9.4  | 11.0  | 6.1–18.3   | 6/2      | 30000 |

Table 8: Same-machine repeat audit for the strict FAISS CPU/bge-small choices. The p99 range is over completed local repeat/configured-client rows and is supplementary evidence, not a hardware-population interval.

| Workload | Rows | Repeats | Quality       | post@5s | p99 min/med/max ms | Errors |
|----------|------|---------|---------------|---------|--------------------|--------|
| S1       | 9    | 3       | nDCG@10=0.506 | –       | 2.4/8.9/27.7       | 0      |
| S2       | 2    | 2       | nDCG@10=0.506 | 0.840   | 21.4/47.8/74.1     | 0      |
| S3       | 9    | 3       | Cov@10=0.852  | –       | 3.6/9.6/20.6       | 0      |

Table 9: Protocol components whose omission changes or weakens deployment conclusions.

| Missing component          | Wrong conclusion caused by omission                                                                       | Evidence                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| p99 latency gate           | S3 cost-only/no-p99 choice shifts to lancedb-inproc/bge-small instead of faiss-cpu/bge-small              | decision table and threshold-sensitivity panel     |
| Objective separation       | Quality-first decisions differ from strict-latency decisions on S1, S2, S3                                | decision table                                     |
| Budget ladder              | B0 screening fails to preserve B2 top-1 decisions for S1, S2                                              | budget-stability figure and decision table         |
| Paired S3 audit            | pgvector/bge-base appears quality-first, but the matched 5,000-query audit removes the substantive margin | S3 paired audit table                              |
| Paired S2 competitor audit | Qdrant does not hide a quality or post-insert retrievability win                                          | S2 competitor audit table                          |
| Conformance gate           | Engines without passing local conformance rows do not enter paper-facing result tables                    | benchmark scope, support table, and artifact audit |

Table 10: B2 decision-surface rows for strict survivors and objective winners. Rows include all strict-p99 survivors plus cost-only/no-p99 and quality-first choices when they are outside the strict survivor set; p99 max is the maximum observed B2 p99 across configured read-client rows and repeats.

| Workload          | Role                            | Engine         | Emb.      | Quality | post@5s | Cost proxy | p99 max | Strict pass |
|-------------------|---------------------------------|----------------|-----------|---------|---------|------------|---------|-------------|
| S1 single-hop     | strict choice; cost-only/no-p99 | faiss-cpu      | bge-small | 0.506   | –       | 262.759    | 11.2    | yes         |
| S1 single-hop     | strict survivor                 | lancedb-inproc | bge-small | 0.506   | –       | 262.759    | 172.1   | yes         |
| S1 single-hop     | strict survivor                 | qdrant         | bge-small | 0.500   | –       | 263.311    | 15.8    | yes         |
| S1 single-hop     | strict survivor                 | faiss-cpu      | bge-base  | 0.527   | –       | 282.561    | 36.9    | yes         |
| S1 single-hop     | strict survivor                 | qdrant         | bge-base  | 0.522   | –       | 283.320    | 16.6    | yes         |
| S1 single-hop     | quality-first                   | lancedb-inproc | bge-base  | 0.527   | –       | 282.561    | 337.4   | no          |
| S2 streaming      | strict choice; cost-only/no-p99 | faiss-cpu      | bge-small | 0.506   | 0.830   | 262.759    | 10.0    | yes         |
| S2 streaming      | strict survivor                 | lancedb-inproc | bge-small | 0.506   | 0.830   | 262.759    | 147.8   | yes         |
| S2 streaming      | strict survivor                 | qdrant         | bge-small | 0.501   | 0.830   | 263.114    | 7.8     | yes         |
| S2 streaming      | strict survivor; quality-first  | faiss-cpu      | bge-base  | 0.527   | 0.846   | 282.561    | 21.7    | yes         |
| S2 streaming      | strict survivor                 | qdrant         | bge-base  | 0.526   | 0.846   | 282.861    | 10.5    | yes         |
| S3 multi-evidence | strict choice                   | faiss-cpu      | bge-small | 0.852   | –       | 129.325    | 22.8    | yes         |
| S3 multi-evidence | strict survivor                 | qdrant         | bge-small | 0.815   | –       | 129.835    | 18.3    | yes         |
| S3 multi-evidence | strict survivor                 | faiss-cpu      | bge-base  | 0.871   | –       | 130.029    | 56.2    | yes         |
| S3 multi-evidence | strict survivor                 | qdrant         | bge-base  | 0.848   | –       | 130.474    | 35.0    | yes         |
| S3 multi-evidence | cost-only/no-p99                | lancedb-inproc | bge-small | 0.847   | –       | 128.492    | 739.6   | no          |
| S3 multi-evidence | quality-first                   | pgvector       | bge-base  | 0.873   | –       | 131.113    | 1217.3  | no          |

Table 11: Strict-latency decision margins against the nearest strict-cost candidate.  $\Delta$  values are candidate minus strict choice;  $\Delta$ p99 is computed from Table 7’s p99-max values.

| Workload          | Strict choice                      | Next strict candidate                   | Metric  | $\Delta$ quality | $\Delta$ cost | $\Delta$ p99 | Interpretation                                              |
|-------------------|------------------------------------|-----------------------------------------|---------|------------------|---------------|--------------|-------------------------------------------------------------|
| S1 single-hop     | faiss-cpu / BAAl/bge-small-en-v1.5 | lancedb-inproc / BAAl/bge-small-en-v1.5 | nDCG@10 | +0.000           | +0.000        | +160.9 ms    | cost/quality tie; p99 tie-break                             |
| S2 streaming      | faiss-cpu / BAAl/bge-small-en-v1.5 | lancedb-inproc / BAAl/bge-small-en-v1.5 | nDCG@10 | +0.000           | +0.000        | +137.8 ms    | cost/quality tie; p99 tie-break                             |
| S3 multi-evidence | faiss-cpu / BAAl/bge-small-en-v1.5 | qdrant / BAAl/bge-small-en-v1.5         | Cov@10  | -0.036           | +0.510        | -4.6 ms      | candidate is more expensive, lower quality, slightly faster |

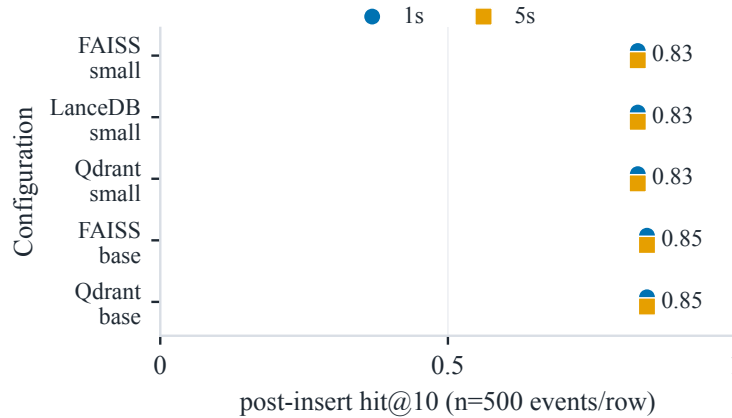


Figure 6: S2 post-insert top-10 retrievability for B2 strict survivors. One- and five-second probes overlap in the archived rows, showing no additional recoveries between the fixed probes.

Table 12: S2 write/retrieve diagnostics for B2 strict survivors and the quality-first row. Probe-window p95 is derived from the fixed 1s/5s post-insert probes after adapter insert plus flush returns; it is not a direct insert-index visibility-latency measurement.

| Role                           | Engine         | Emb.      | nDCG@10 | post@1s | post@5s | probe-window p95 | Events | Errors |
|--------------------------------|----------------|-----------|---------|---------|---------|------------------|--------|--------|
| strict choice                  | faiss-cpu      | bge-small | 0.506   | 0.830   | 0.830   | 5000.0 ms        | 500    | 0      |
| strict survivor                | lancedb-inproc | bge-small | 0.506   | 0.830   | 0.830   | 5000.0 ms        | 500    | 0      |
| strict survivor                | qdrant         | bge-base  | 0.526   | 0.846   | 0.846   | 5000.0 ms        | 500    | 0      |
| strict survivor                | qdrant         | bge-small | 0.501   | 0.830   | 0.830   | 5000.0 ms        | 500    | 0      |
| strict survivor; quality-first | faiss-cpu      | bge-base  | 0.527   | 0.846   | 0.846   | 5000.0 ms        | 500    | 0      |

Table 13: S2 strict-choice post-insert example outcomes. Counts are archived freshness observations for FAISS CPU/bge-small B2 rows; examples show IDs only because the archived observation schema records event/query/document IDs and hit flags, not ranked-list text.

| Outcome                     | Count | Rate  | Example event (vis.) | Query             | Target doc       | Hits 1s/5s | Interpretation                                                                        |
|-----------------------------|-------|-------|----------------------|-------------------|------------------|------------|---------------------------------------------------------------------------------------|
| retrieved by 1s             | 830   | 0.830 | 0 (<0.1 ms)          | 7bb29eb4...3c8962 | 7bb29eb4...p0_c0 | 1/1        | target passage appears in top-10 at the first fixed probe                             |
| recovered between 1s and 5s | 0     | 0.000 | –                    | –                 | –                | –/–        | no archived strict-choice events recovered only at the later probe                    |
| missed by 5s                | 170   | 0.170 | 6 (5000.0 ms)        | 31f7aad...4ba2ac  | 31f7aad...p0_c0  | 0/0        | target passage absent from top-10 at the final fixed probe; outcome is censored at 5s |
| inconsistent archived flags | 0     | 0.000 | –                    | –                 | –                | –/–        | archived hit flags do not form a monotone probe outcome                               |

Table 14: Conformance and behavior-card support for paper-facing reportability.

| Engine          | Behavior card | Conformance | Reported? | Exclusion reason        |
|-----------------|---------------|-------------|-----------|-------------------------|
| faiss-cpu       | faiss_cpu.md  | pass        | yes       |                         |
| lancedb-inproc  | lancedb.md    | pass        | yes       |                         |
| lancedb-service | lancedb.md    |             | no        | missing conformance row |
| pgvector        | pgvector.md   | pass        | yes       |                         |
| qdrant          | qdrant.md     | pass        | yes       |                         |

Table 15: Engine version and run-mode metadata for the reported matrix.

| Engine          | Mode            | Version                                                                    | Metric | Dim.     | Reported? |
|-----------------|-----------------|----------------------------------------------------------------------------|--------|----------|-----------|
| faiss-cpu       | in-process      | faiss-cpu 1.8.0.post1; archived engine_version=cpu                         | ip     | 384, 768 | yes       |
| lancedb-inproc  | in-process      | lancedb 0.25.3; archived engine_version=local                              | ip     | 384, 768 | yes       |
| lancedb-service | service API     | not reported; no passing local lancedb-service conformance row             | ip     |          | no        |
| pgvector        | server database | pgvector 0.8.2 on PostgreSQL 16; image pgvector/pgvector:0.8.2-pg16-trixie | ip     | 384, 768 | yes       |
| qdrant          | server API      | Qdrant server v1.17.1; image qdrant/qdrant:v1.17.1; requests 2.31.0        | ip     | 384, 768 | yes       |

Table 16: Engine index/search and flush semantics for the reported matrix.

| Engine          | Index/search                                                                                                                         | Process model                                    | Flush/commit                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------|
| faiss-cpu       | index_type=flat (exact IndexFlatIP); search_sweep recorded but ignored by flat index; top_k=10                                       | Python process-local adapter                     | staged writes become visible after flush_or_commit rebuild/update |
| lancedb-inproc  | default LanceDB table search; adapter does not create a secondary index; search_sweep recorded but not consumed by adapter; top_k=10 | Python in-process library with local LanceDB URI | staged writes become visible after flush_or_commit                |
| lancedb-service | default adapter configuration                                                                                                        | external HTTP service contract                   | depends on service implementation; no passing local row           |
| pgvector        | index_method=ivfflat (adapter default); search_sweep=[{"ivfflat_probes": 32}, {"ivfflat_probes": 64}]; top_k=10                      | PostgreSQL server plus Python adapter            | writer transaction commit; readers observe committed state        |
| qdrant          | search_sweep=[{"hmsw_cf": 32}, {"hmsw_cf": 64}]; top_k=10                                                                            | Qdrant server plus Python adapter                | upsert/update/delete requests use wait=true                       |

Table 17: Matched-query S3 quality audit for the quality-first margin. Each row evaluates the same 5,000-query HOTPOTQA-MAXIONBENCH matched-audit subset once with BAAI/bge-base-en-v1.5 and clients\_read=4. Paired deltas are against FAISS exact FlatIP in evidence\_coverage@10; intervals are paired bootstrap 95% intervals.

| Engine / search          | evidence_coverage@10 | p99 (ms) | QPS   | Paired $\Delta$ evidence_coverage@10 |
|--------------------------|----------------------|----------|-------|--------------------------------------|
| FAISS CPU / exact FlatIP | 0.8712               | 17.6     | 247.6 | reference                            |
| pgvector / IVF32         | 0.8711               | 306.0    | 15.2  | −0.0001 [−0.0003, 0.0000]            |
| pgvector / IVF64         | 0.8711               | 346.2    | 14.6  | −0.0001 [−0.0003, 0.0000]            |

Table 18: S3 all-evidence hit@10 audit rows. All-hit observation rows count only rows used to compute the stricter binary all-hit metric; rows with Cov@10 and zero all-hit observations inherit Cov@10 from the decision table but have no archived all-hit audit sample.

| Role                            | Engine         | Embedding | Cov@10 | All-hit@10 | All-hit obs. rows |
|---------------------------------|----------------|-----------|--------|------------|-------------------|
| strict choice                   | faiss-cpu      | bge-small | 0.852  | 0.722      | 61896             |
| quality-first archived row      | pgvector       | bge-base  | 0.873  | —          | 0                 |
| cost-only/no-p99 row            | lancedb-inproc | bge-small | 0.847  | —          | 0                 |
| high-quality strict-latency row | faiss-cpu      | bge-base  | 0.871  | —          | 0                 |
| next strict-cost candidate      | qdrant         | bge-small | 0.815  | —          | 0                 |

Table 19: Matched S1/S2 FAISS CPU versus LanceDB in-process audit with BAAI/bge-small-en-v1.5. Paired deltas are LanceDB minus FAISS.

| Workload | Comparison      | Quality obs. | $\Delta$ quality | 95% CI           | Post-insert events | $\Delta$ post_insert_hit@10,5s | p99 direction   | Conclusion                                                    |
|----------|-----------------|--------------|------------------|------------------|--------------------|--------------------------------|-----------------|---------------------------------------------------------------|
| S1       | LanceDB – FAISS | 1788         | 0.0000           | [0.0000, 0.0000] | —                  | —                              | FAISS lower p99 | tail-latency tie-break; paired quality interval contains zero |
| S2       | LanceDB – FAISS | 1788         | 0.0000           | [0.0000, 0.0000] | 40                 | 0.0000                         | FAISS lower p99 | tail-latency tie-break; paired quality interval contains zero |

Table 20: S2 larger same-machine competitor check with BAAI/bge-small-en-v1.5, clients\_read=8, clients\_write=2, two repeats per engine, 1,788 matched quality queries, and 200 matched post-insert events. Paired deltas are Qdrant minus FAISS with 95% paired bootstrap intervals.

| Engine / search  | nDCG@10 | Paired $\Delta$ nDCG@10    | post_insert_hit@10,5s | Mean / max p99 (ms) | Errors |
|------------------|---------|----------------------------|-----------------------|---------------------|--------|
| FAISS CPU / ef64 | 0.5055  | reference                  | 0.840                 | 47.8 / 74.1         | 0      |
| Qdrant / ef64    | 0.5028  | −0.0027 [−0.0052, −0.0006] | 0.840                 | 17.1 / 20.7         | 0      |

Table 21: Quality floors and strict-p99 B2 survivors used by the strict-latency decision rule. Floor values are read from the archived quality\_target configuration key and match the report generator’s floor rule. S2 post-insert metrics are diagnostics in the archived rule, not additional hard gates.

| Workload          | Quality metric                     | Quality floor source               | Floor value | Strict-p99 survivors |
|-------------------|------------------------------------|------------------------------------|-------------|----------------------|
| S1 single-hop     | nDCG@10                            | archived config key quality_target | 0.25        | 5                    |
| S2 streaming      | nDCG@10 plus post-insert reporting | archived config key quality_target | 0.25        | 5                    |
| S3 multi-evidence | evidence_coverage@10               | archived config key quality_target | 0.30        | 4                    |



## C Claim-evidence map

- Claim: MaxionBench is conformance-gated. Evidence: Table 14 reports passing conformance and behavior cards for all included engines, and the report excludes LanceDB service because conformance is missing.
- Claim: the paper contributes a reusable decision-audit protocol rather than only a benchmark implementation. Evidence: the decision-card protocol in Section 2.3 defines the conformance gate, workload evaluation, objective-specific ranking, budget-stability check, paired audit, and emitted card fields.
- Claim: strict 200 ms max-row p99 deployment selects FAISS CPU with BAAI/bge-small-en-v1.5 for S1, S2, and S3. Evidence: Table 2 and Table 3.
- Claim: supplementary same-machine repeat evidence does not contradict the strict FAISS CPU/bge-small decisions. Evidence: Table 8 summarizes completed local repeat rows for S1, S2, and S3.
- Claim: the selected rows are not cherry-picked from a hidden surface. Evidence: Figure 2 and Table 10 show all B2 strict survivors plus objective winners outside the strict survivor set.
- Claim: strict decisions have different margins across workloads. Evidence: Table 11 shows S1/S2 as cost/quality ties resolved by latency and S3 as lower-cost and higher-quality than the next strict-cost candidate.
- Claim: objective choice changes deployment conclusions. Evidence: Table 3 shows different quality-first winners for all workloads and a cost-only/no-p99 shift for S3.
- Claim: the S3 quality-first pgvector result is indistinguishable from FAISS under matched queries. Evidence: Figure 4 and Table 17 report a matched-query audit over the 5,000-query S3 matched-audit subset in which pgvector trails the FAISS exact FlatIP baseline by 0.0001 evidence\_coverage@10 with paired 95% interval [-0.0003, 0.0000].
- Claim: the S2 strict decision is not hiding a Qdrant quality or post-insert retrievability win. Evidence: Table 12 summarizes B2 post-insert diagnostics, Table 13 gives archived strict-choice success/miss examples and counts, and Table 20 reports a larger same-machine paired check with 1,788 matched quality observations and 200 matched post-insert events.
- Claim: budget stability is workload-dependent. Evidence: Figure 5; B0-to-B2 top-1 agreement is 0 for S1/S2 and 1 for S3, with Spearman correlations 0.79, 0.43, and 0.31.

Table 22: Reviewer-facing evidence map. The main claims are framed as benchmark and decision-audit claims rather than universal engine rankings.

| Claim                                                         | Main evidence                                                                   | Residual risk                                                           | Paper stance                                 |
|---------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------|
| Conformance-gated reporting                                   | 72 strict-schema run dirs; support table                                        | Adapter coverage is local                                               | Supported protocol claim                     |
| Strict-latency decisions                                      | Figure 2; main and decision tables; margin table                                | One hardware profile                                                    | Local deployment evidence                    |
| Strict FAISS repeat evidence                                  | Same-machine repeat audit in Table 8                                            | Repeats are local and not hardware-population intervals                 | Supplementary robustness check               |
| Decision surface is visible                                   | B2 strict survivors and objective rows in Figure 2 and Table 10                 | Full matrix remains appendix/artifact material                          | Anti-cherry-picking evidence                 |
| S2 has no hidden QoQnt quality/post-insert retrievability win | S2 diagnostics in Tables 12-13; larger same-machine paired S2 check in Table 20 | Post-insert probes remain capped and observations lack ranked-list text | Sanity check, not universal latency evidence |
| S3 pvector quality edge is not substantive                    | 5,000-query paired audit against FAISS exact FlatIP in Figure 4 and Table 17    | No high-eff/probe service sweep beyond 64                               | Objective sensitivity, not quality win       |
| Budget stability is workload-dependent                        | Figure 5; B0-to-B2 top-1 and Spearman stats                                     | Lower ranks remain noisy                                                | Report budgets explicitly                    |

## NeurIPS Paper Checklist

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes].

Justification: The abstract and introduction claim a conformance-gated local decision audit and explicitly limit the scope to single-node deployment evidence; Figure 2, Tables 2 and 3, and the appendix claim-evidence map provide the supporting evidence.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes].

Justification: The manuscript includes a dedicated limitations section describing the single-node setting, absence of second-machine/distributed/GPU claims, bounded S3 corpus, S2 post-insert retrievability interpretation, service-sweep limits, and budget-stability limits.

### 3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

Answer: [N/A].

Justification: The paper does not present theoretical results; it defines benchmark metrics and decision rules.

### 4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes].

Justification: The appendix and artifact README list the workflow commands, archive identifier, validation command, runtime metadata, checksums, conformance matrix, generated report artifacts, and local artifact-readiness verifier.

### 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

Answer: [Yes].

Justification: The benchmark code, commands, generated manifests, and archived result bundle are part of the repository prepared with this manuscript; any public release should preserve anonymization at submission time.

### 6. Experimental setting/details

Question: Does the paper specify all training and test details necessary to understand the results?

Answer: [Yes].

Justification: The benchmark-design section specifies engines, datasets, embedding tiers, concurrency pins, budget levels, metrics, decision rules, and conformance gates.

### 7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about statistical significance?

Answer: [Yes].

Justification: The main table reports 95% confidence intervals; targeted S2/S3 audits report paired bootstrap intervals for the highest-risk margins, and Figure 4 visualizes the S3 paired interval.

#### 8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on compute resources needed to reproduce the experiments?

Answer: [Yes].

Justification: The scope and reproducibility sections state the single-node commodity-hardware setting and exact budget ladder; the archived run metadata records per-run resource fields.

#### 9. Code of ethics

Question: Does the research conducted in the paper conform with the NeurIPS Code of Ethics?

Answer: [Yes].

Justification: The work evaluates retrieval infrastructure using public benchmark artifacts and does not involve human subjects, private data, or model training on sensitive data.

#### 10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes].

Justification: The limitations and broader-impact section discusses the benefit of transparent deployment evidence and the risk of overgeneralizing local benchmark winners.

#### 11. Safeguards

Question: Does the paper describe safeguards for responsible release of data or models that have a high risk for misuse?

Answer: [N/A].

Justification: The paper does not release new generative models or scraped high-risk datasets; it uses existing retrieval benchmark assets.

#### 12. Licenses for existing assets

Question: Are creators or original owners of assets properly credited and are licenses and terms of use explicitly mentioned and respected?

Answer: [Yes].

Justification: The manuscript cites the datasets, software systems, and embedding models used; final submission should include any repository-level license manifests alongside the artifacts.

#### 13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes].

Justification: HOTPOTQA-MAXIONBENCH is documented through its preprocessing description, manifest, checksums, archived bundle, and Croissant-style metadata included with the reviewer artifact.

#### 14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include instructions, screenshots, and compensation details?

Answer: [N/A].

505 Justification: The paper does not involve crowdsourcing or human-subject experiments.  
506 **15. Institutional review board approvals or equivalent for research with human subjects**  
507 Question: Does the paper describe risks to study participants and IRB approvals or equivalent?  
508 Answer: [N/A].  
509 Justification: The paper does not involve human-subject research.  
510 **16. Declaration of LLM usage**  
511 Question: Does the paper describe the usage of LLMs if it is an important, original, or  
512 non-standard component of the core methods?  
513 Answer: [N/A].  
514 Justification: LLMs are not an important, original, or non-standard component of the  
515 benchmark method; the benchmark evaluates retrieval infrastructure and local embedding  
516 models.  
517